

Problem - 1

$$X = [5.4, 3.0, 1.5, 0.2]$$

$$w_1 = 1, w_2 = 3, w_3 = 5, w_4 = 7$$

$$b = 1.5$$

$$Z = x_1 w_1 + x_2 w_2 + x_3 w_3 + x_4 w_4 + b$$

Solution

$$Z = (5.4 \times 1) + (3.0 \times 3) + (1.5 \times 5) + (0.2 \times 7) + 1.5$$

$$= 5.4 + 9 + 7.5 + 1.4 + 1.5$$

$$\boxed{Z = 24.8}$$

Apply sigmoid

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$= \frac{1}{1 + e^{-24.8}}$$

$$\sigma(z) \approx 1$$

$$\boxed{h=1}$$

output layer

$$w = 1.2, b = 0.1$$

$$Z_{\text{setosa}} = h(1.2) + 0.1$$

$$= 1(1.2) + 0.1$$

$$\boxed{Z_{\text{setosa}} = 1.3}$$

Sigmoid:

$$\sigma(1.3) = \frac{1}{1 + e^{-1.3}} = \underline{\underline{0.786}}$$

Versicolor

$$w = 1.5, b = 0.2$$

$$Z_{\text{versicolor}} = 1(1.5) + 0.2$$

$$\boxed{1.7}$$

$$\sigma(1.7) \approx \boxed{0.846}$$

Virginica

$$w = 1.8, b = 0.1$$

$$Z_{\text{virginica}} = 1(1.8) + 0.1$$

$$\boxed{1.9}$$

$$\sigma(1.9) \approx 0.870$$